

premium puzzle—that equity returns are high relative to this highly stylized, artificial, Mehra–Prescott economy—purely academic, or does it have relevance for how we invest assets in the real world?

The equity premium puzzle is important because many investors hold large amounts of equities based on the historical record of high returns. Will such returns persist going forward, as in our motivating example, or will equity returns permanently enter a new phase that looks more like an enduring version of the 2000s’ Lost Decade? Understanding the reasons for the equity premium provides a rational basis for deciding whether we should continue to hold lots of equities in the future, rather than simply continuing to do so because that’s what we have done in the past. While economists have developed many explanations, I cover four that are most relevant for investors.

3. Explanations for the Equity Premium Puzzle

3.1. MARKET RISK AVERSION IS (SOMETIMES) VERY HIGH

The first resolution to the equity premium puzzle is that the market’s risk aversion is extremely high. It can be shown that a lower bound for the risk aversion of the market, γ , or the representative agent’s risk aversion, is given by

$$\gamma > \frac{\text{Sharpe Ratio}}{\sigma_c}, \quad (8.1)$$

where *Sharpe Ratio* denotes the Sharpe Ratio of the equity market portfolio and σ_c is the volatility of real consumption growth.³

The lower bounds for risk aversion given by the data in Table 8.3 are approximately 20. These are much larger than the upper bounds of “reasonable” risk aversion levels of 10 claimed by Mehra and Prescott and discussed in chapter 2. If we compute the implied risk aversion levels exactly assuming log-normal consumption growth, then risk aversion of the average investor to match the equity premium is above 120. Estimates like these are pervasive throughout the literature beginning with the seminal work of Hansen and Singleton (1983).

Maybe the representative agent’s risk aversion is high, perhaps even above 120. This is problematic for the obvious reason that individuals do not have risk aversion that high. There is a second reason labeled the *risk-free rate puzzle* due to Philippe Weil (1989). While very high risk aversion can match the equity premiums in data, high risk aversions also raise the risk-free rate in the Mehra–Prescott model. A very risk-averse representative agent would demand a high premium to

³ See Cochrane (2001) for a derivation. This is a special case of a more general bound on the pricing kernel (see chapter 6) developed by Hansen and Jagannathan (1991). Lars Hansen received his Nobel Prize in 2013.